A Computer-Assisted All-Orders Certificate for a Six-Field Flux Subsystem on a Calabi–Yau Threefold

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Abstract

We develop a computer-assisted framework for proving existence and local uniqueness of a supersymmetric flux-sector root on an explicit toric Calabi–Yau threefold with $h^{2,1} = 5$ and $h^{1,1} = 81$. The system contains the axiodilaton and five complex-structure fields. A real 12×12 Krawczyk operator certifies a root of a finite instanton system, while exact Chow arithmetic and geometric-series majorants control the omitted GKZ degrees through periods, F-terms, and their Jacobian. The finite 557-degree source sector is included in a recentered Arb evaluator with a box-wide interval enclosure, and an independent verifier checks the complete degree partition and the finite-plus-infinite Krawczyk inclusion. Five light quotient classes are identified as persistent rigid rational complete-intersection curves, as a supporting geometric result rather than part of the central proof.

1 Context and contribution

Algorithmic algebraic geometry and numerical homotopy continuation have long been used to locate and enumerate flux vacua [1, 2, 3]. More recent work has made large-modulus mirror symmetry computationally effective even for Calabi–Yau threefolds with many moduli [4], while interval Krawczyk operators provide practical certificates for isolated numerical solutions [5]. The explicit geometry and flux choice used here descend from the ancillary data of [6]. Exact flux landscapes can also arise on special symmetry loci [7]. Recent analyses stress that instanton corrections can change the location and even the multiplicity of flux solutions near large-complex-structure boundaries [8].

The contribution of this paper is to join these ingredients into one end-to-end certificate. Every degree is assigned either to an exact finite Chow computation or to an analytic infinite-tail majorant; the resulting period bounds are propagated through the mirror map, the six F-terms, their real Jacobian, and the final Krawczyk operator. To our knowledge, the cited flux-vacuum literature does not provide this combination of an explicit multi-parameter compact Calabi–Yau model, all-orders GKZ accounting, and an interval proof of local uniqueness. This is a deliberately narrower claim than a proof of a complete physical vacuum.

2 Claim and scope

Let $F : \mathbb{R}^{12} \to \mathbb{R}^{12}$ be the real form of the six complex equations $D_I W = 0$, with I running over the axiodilaton and five complex-structure moduli. The intended result is strictly local.

Certified quantity	Value
Complex fields / real equations	6 / 12
Finite source rows	557
Low-total representations	8150
Partition $Z/S/M/U$	7150/527/21/452
Box radius r	1.000000×10^{-30}
$\|CF(x_0)\|_\infty$	3.727099×10^{-69}
$\|I - CJ(X)\|_\infty$	7.573429×10^{-4}
Finite Krawczyk image	7.573429×10^{-34}
Finite strict margin	9.992427×10^{-31}
$\|C_{\text{new}}\|_\infty$	1.446862×10^{23}
Maximum all-orders addition	1.081536×10^{-53}
Tail / finite margin	1.082356×10^{-23}
Strict real components	12
Authenticated provenance files	16

Table 1: Machine-generated summary of the admitted certificate. The displayed decimals are derived from exact rational bounds.

Theorem 1 (Certified six-field local theorem)**.** *For the flux and basis data specified in the archived artifact, the complete closed-sector GKZ system has exactly one zero in the declared componentwise box of radius* 10^{-30} *around the archived center.*

The theorem does not claim global uniqueness, simultaneous stabilization of the 81 Kähler moduli or open-string fields, uplifted stability, or a complete physical vacuum.

3 Model and equations

The compactification is the toric hypersurface dataset `5-81-3213` of [6], with $(h^{2,1}, h^{1,1}) = (5, 81)$. We use the primitive integral divisor basis

$$(J_1, J_2, J_3, J_4, J_5) = (D_2, D_3, D_4, D_5, D_9), \qquad \int_X c_2(X) \wedge J_a = (16, 8, 12, 20, 8)_a. \tag{1}$$

The independent Chow reconstruction has full column rank over $\mathbb{Q}$; the alternative compatible divisor tuple is related to the displayed one by a unimodular matrix. In the displayed basis, the nonzero symmetric triple intersections are

$$(\kappa_{111}, \kappa_{112}, \kappa_{114}, \kappa_{122}, \kappa_{124}, \kappa_{134}, \kappa_{144}, \kappa_{222}) = (-2, 2, -2, -2, 2, 2, -2, 2), \tag{2}$$
$$(\kappa_{224}, \kappa_{244}, \kappa_{334}, \kappa_{344}, \kappa_{444}, \kappa_{445}, \kappa_{455}, \kappa_{555}) = (-2, 2, -2, 2, -4, 2, -2, 2). \tag{3}$$

Let $\Phi = (\tau, z^1, \ldots, z^5)$, where τ is the axiodilaton and the z^a are complex-structure coordinates. The integral flux vectors, in the archived symplectic ordering, are

$$f = (-2, 4, -10, -2, 3, -3, 0, 3, -5, 2, -2, -5), \tag{4}$$
$$h = (0, -5, 5, -4, -1, 5, 0, 0, 0, 0, 0, 0). \tag{5}$$

Writing $f = (f_B, f_A)$ and $h = (h_B, h_A)$ in two six-component blocks and $\Pi = (X^I, F_I)$ with $X^0 = 1$ and $X^a = z^a$, our convention is

$$W = \sqrt{\frac{2}{\pi}} \left[(f_B - \tau h_B)^T X - (f_A - \tau h_A)^T F \right], \qquad \mathcal{F}_I = D_I W. \tag{6}$$

The real system is ordered as

$$x = (\operatorname{Re}\tau, \operatorname{Re}z^1, \ldots, \operatorname{Re}z^5, \operatorname{Im}\tau, \operatorname{Im}z^1, \ldots, \operatorname{Im}z^5). \tag{7}$$

Its recentered witness begins

$$\begin{aligned} x_0 \simeq (&-12, -19.5, -29.5, -15, -15, -5; \\ &19.81991494, 32.20736178, 48.72395756, 24.77489367, 24.77489367, 8.258297891), \end{aligned} \tag{8}$$

with the full decimal center serialized in the certificate.

In the large-complex-structure chart the genus-zero prepotential is written as

$$\mathcal{G}(z) = -\frac{1}{6}\kappa_{abc}z^a z^b z^c + \frac{1}{2}A_{ab}z^a z^b + \frac{1}{24}c_{2,a}z^a + \frac{\chi\,\zeta(3)}{2(2\pi i)^3} - \frac{1}{(2\pi i)^3}\sum_{\beta>0} n_\beta^{(0)}\operatorname{Li}_3(e^{2\pi i\beta\cdot z}), \tag{9}$$

where $\chi = -152$, and A_{ab} is the integral quadratic convention fixed by the archived symplectic basis. The period blocks used above are $X = (1, z^a)$ and $F = (2\mathcal{G} - z^a\mathcal{G}_a, \mathcal{G}_a)$.

4 Finite-system interval certificate

For a center x_0, interval box $X = x_0 + [-r, r]^{12}$, and numerical inverse witness C, define

$$K(X) = x_0 - CF(x_0) + (I - CJ(X))(X - x_0). \tag{10}$$

The Arb calculation reports the conservative scalar bounds

$$r = 10^{-30}, \qquad \|CF(x_0)\|_\infty < 3.73 \times 10^{-69}, \tag{11}$$

$$\|I - CJ(X)\|_\infty < 7.58 \times 10^{-4}. \tag{12}$$

Consequently the finite-system image radius is bounded by

$$3.73 \times 10^{-69} + 7.58 \times 10^{-4} r < r. \tag{13}$$

This proves existence and local uniqueness for the implemented finite system. The attachment also contains all twelve componentwise inclusions; the scalar relaxation and the exact proof summary are reported in Table 1.

5 GKZ degree decomposition

The secondary-cone monoid is decomposed into a zero-action direction and five positive-action Hilbert generators. Through transverse total eight, 8150 representations are classified as shown in Table 2.

The 452 representations reduce to 188 distinct nonzero Chow charges. Exact rational coefficient norms are used for this low sector. All totals at least nine enter a geometric majorant, whose action is bounded below by $66.8922\ldots$. The resulting uncovered-sector bound on a second logarithmic period derivative is 1.743×10^{-77}.

6 The finite-source-sector admission lemma

Proposition 1 (Certified finite-sector lemma)**.** *On a declared complex neighborhood containing the final Krawczyk box, the difference between the complete 557-degree Chow-reduced period jet and the finite baseline used in the interval solve has rigorous componentwise bounds through derivative order two. Those bounds are propagated through the mirror map, symplectic periods, F-terms, and real Jacobian.*

Category	Representations
Chow zero	7150
Finite source table	527
Displayed primitive multicover	21
Uncovered nonzero Chow sector	452
Total	8150

Table 2: Low-total degree ledger. The artifact contains the explicit disjoint membership lists and their canonical digest.

The full finite system shifts the original four-term center by approximately 2.93×10^{-20}, so the old radius-10^{-30} box is not retained. The complete finite sector is instead included in the Arb evaluator and certified in a new radius-10^{-30} box around the recentered root. The inverse mirror map is simultaneously certified by a five-component contraction argument.

7 Certified all-orders perturbation

For the already covered complement of the finite source sector, propagation gives

$$\|\Delta F\|_\infty < 8.543 \times 10^{-77}, \tag{14}$$

$$\|\Delta J\|_\infty < 2.810 \times 10^{-73}, \tag{15}$$

$$\max_i \rho_i < 1.082 \times 10^{-53}. \tag{16}$$

Here the final verifier computes, using rational arithmetic,

$$\rho_i = \sum_{j=1}^{12} |C_{\mathrm{new},ij}| \big(\Delta F_j + r\Delta J_j\big). \tag{17}$$

All 144 upper bounds for $|C_{\mathrm{new},ij}|$ are serialized, and their exact row sums give $\|C_{\mathrm{new}}\|_\infty < 1.447 \times 10^{23}$. Together with Proposition 1 and the exhaustive membership ledger, these inequalities give the stated all-orders six-field theorem.

Proof of Theorem 1. Let $X = x_0 + [-r, r]^{12}$. The Arb witness encloses the complete finite 557-row system and its Jacobian on X, including the inverse mirror map, and gives

$$\|CF_{\mathrm{fin}}(x_0)\|_\infty + \|I - CJ_{\mathrm{fin}}(X)\|_\infty r \le 7.574 \times 10^{-34}. \tag{18}$$

The explicit low-total ledger and the high-total geometric majorant exhaust the complementary GKZ degrees. Their value and derivative bounds hold on the radius-0.1 logarithmic domain, which contains X after recentering. Exact propagation through C_{new} adds at most 1.082×10^{-53} to each real component of the Krawczyk image. Hence

$$K_{\mathrm{full}}(X) \subset \operatorname{int} X \tag{19}$$

in all twelve components. The Krawczyk theorem therefore gives a zero of the full closed-sector map in X and uniqueness within the declared box. $\square$

8 Five supporting rigid curves

Five minimum-volume quotient-basis classes have unique representatives among the 3570 pairs of prime toric divisors:

$$D_{15}D_{24}X, \quad D_{13}D_{84}X, \quad D_{9}D_{81}X, \quad D_{10}D_{80}X, \quad D_{13}D_{82}X. \tag{20}$$

For each representative, exact intersection data and face combinatorics give a primitive genus-zero class with normal bundle $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$. The restricted hypersurface equation is a primitive binomial whose coefficients remain nonzero in the all-orders logarithmic coefficient box, so the curves persist there. The interpretation of these classes follows the toric complete-intersection analysis of [4].

This statement is intentionally not exhaustive. The complete-intersection audit contains 54 non-potent divisor-pair records representing 52 distinct full classes. The five curves support the geometric interpretation of the selected quotient basis but do not prove an all-orders GV vanishing theorem.

9 Reproducibility and trusted base

The release contains a small Python verifier with no third-party dependency. It recomputes strict scalar and componentwise inclusion inequalities, checks scope constraints, authenticates upstream artifacts by SHA-256, and fails if the finite-sector or degree-partition witnesses are absent. The heavy generators use Arb, exact rational Chow arithmetic, Normaliz, and CYTools. Approximate roots and preconditioners are witnesses; interval inclusion is the proof.

Negative tests alter preconditioner entries, tail bounds, one radius, one category count, the local-uniqueness scope, and the geometric exhaustion flag. Each alteration must invalidate the certificate. A separate dependency-free verifier recomputes the twelve tail contributions and the final inclusion inequality using exact rational arithmetic.

The table included in the manuscript is generated directly from `data/certificate.json` and `data/recentered_tail_propagation.json`. It is therefore a build artifact rather than a second manually maintained statement of the numerical claim.

The exact public release used for this article is version v1.0.3, archived under the Zenodo concept DOI (accessed 12 August 2026); the source is maintained in the public GitHub repository.

10 Discussion

Computer-assisted proofs can bridge the gap between formal instanton expansions and numerical flux vacua, but only when every series sector is accounted for by a disjoint and exhaustive certificate. The present workflow provides a strict enclosure of the complete finite source sector, a sharp majorant for the uncovered infinite tail, and an exact componentwise propagation of that tail through the recentered preconditioner. Their strict combined inclusion proves the stated local six-field theorem. No claim about the full physical vacuum is made by this focused result.

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